

# Chasing the Stability Island

A double pendulum driven from regular swinging into chaos:  
problem statement and a three-session laboratory plan

Energy enters the pendulum in one way only. We lift the upper rod to an angle  $\theta_{1,0}$ , let the lower rod hang freely so that  $\theta_{2,0} = 0$ , and release both from rest. Small angles give regular, predictable swinging; large angles give chaos. The question is what happens in between, and whether the boundary is a simple line.

**Equipment.** Camera at 120 fps, 30 s per take. Software: `dp_gui.py` (video tracking), `dp_sim.py` (simulation), `dp_search.py` (angle sweep), `dp_flipmap.py` (the map).

## 1 What we are trying to find out

The criterion of chaos is deliberately crude and easy to see on video: **TTF**, the *time to the first flip* — the moment when the lower rod first goes over the top of its own axle. If that never happens inside our 30 s recording, the release angle counts as regular.

The model makes three predictions worth testing.

- Below  $87.3^\circ$  no flip is possible at all. This is pure energy bookkeeping, not dynamics.
- Above roughly  $130^\circ$  the first flip comes within about 1.5 s, every time.
- Between them the picture is ragged, and there is a band near  $124^\circ$ – $128^\circ$  where no flip occurs in 30 s — an island of calm inside the chaotic sea. *Catching this island is the point of the experiment.*

The island exists only because of friction: without friction the same release angles flip within 2–5 s. That makes it a sharp test of the whole chain — measure friction on the bench, put it into the equations, predict a band that a frictionless textbook model would never show, then find it.

**Why friction cannot be ignored.** Each rod on its own loses half its amplitude in about 30 s, and the assembled pendulum loses roughly half its energy over the same 30 s, which is exactly the length of one recording. A conservative model would be wrong by an amount comparable to the effect we are studying, so viscous friction is part of the model from the start.

## 2 The pendulum and its parameters

### 2.1 Measured directly

### 2.2 Computed from it

A rod hanging from its own hinge is a physical pendulum, so its period gives the moment of inertia about that hinge; subtracting the parallel-axis term gives the moment about the centre

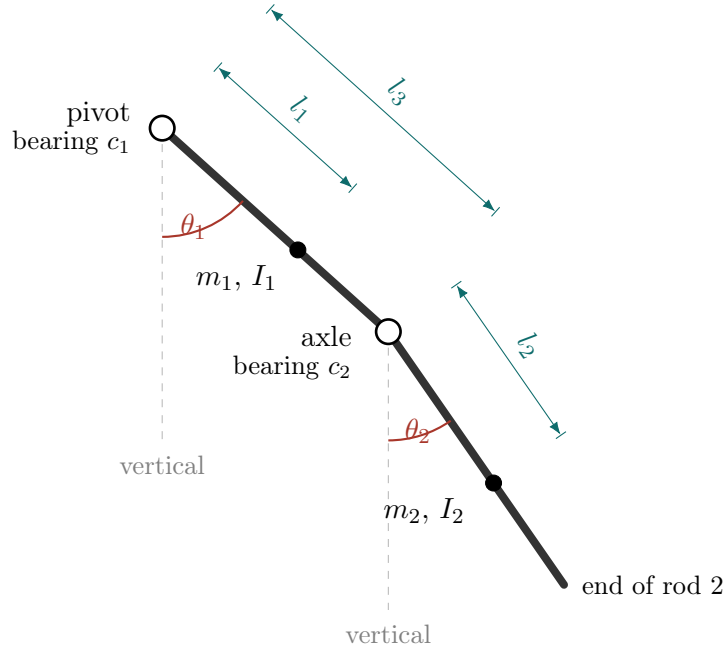


Figure 1: The rig and every symbol used below. Both angles are counted from the downward vertical, positive to the right.  $l_3$  is measured between the two hinges;  $l_1$  and  $l_2$  reach from a hinge to the centre of mass of the rod that hangs from it. Friction lives in the two bearings, with coefficients  $c_1$  at the pivot and  $c_2$  at the axle.

| Symbol     | Quantity                                  | How                     | Typical             |
|------------|-------------------------------------------|-------------------------|---------------------|
| $m_1, m_2$ | mass of each rod, with its bearing parts  | scale, 1 g resolution   | 0.438 kg / 0.480 kg |
| $l_1, l_2$ | hinge to centre of mass of that rod       | balance on a knife edge | 0.215 m / 0.215 m   |
| $l_3$      | pivot to the second axle                  | caliper or steel rule   | 0.255 m             |
| $T_1, T_2$ | period of small swings, rod hanging alone | 20 swings, 5 repeats    | 1.00 s / 1.00 s     |
| $t_{1/2}$  | time for that rod's amplitude to halve    | video, fit the envelope | 30 s / 30 s         |

Table 1: Everything is measured with one rod at a time. That is the only way to separate the two bearings.

of mass. Its decay gives the friction of that one bearing:

$$J = m g l \left( \frac{T}{2\pi} \right)^2 \quad \text{moment of inertia about the hinge} \quad (1)$$

$$I = J - m l^2 \quad \text{about the centre of mass} \quad (2)$$

$$\tau = \frac{t_{1/2}}{2 \ln 2} = \frac{t_{1/2}}{1.386} \quad \text{amplitude decays as } e^{-t/2\tau} \quad (3)$$

$$c = \frac{J}{\tau} \quad \text{viscous coefficient of that bearing} \quad (4)$$

**The one easy mistake.**  $\tau$  is not the half-life. The amplitude halves after  $1.386 \tau$  and the energy halves after  $0.693 \tau$ . Write down what you actually watched — amplitude or energy — before converting.

With the numbers of Table 1 this gives  $I_1 = 0.00315 \text{ kg m}^2$ ,  $I_2 = 0.00346 \text{ kg m}^2$ ,  $\tau_1 = \tau_2 = 21.6 \text{ s}$ ,  $c_1 = 0.00108 \text{ N m s}$  and  $c_2 = 0.00119 \text{ N m s}$ .

### 3 Equations of motion

Both rods are extended bodies, not point masses, so the model keeps their moments of inertia. Five constants absorb all the geometry:

$$a = I_1 + m_1 l_1^2 + m_2 l_3^2 \quad \text{effective inertia of } \theta_1 \quad (5)$$

$$b = m_2 l_3 l_2 \quad \text{coupling between the rods} \quad (6)$$

$$d = I_2 + m_2 l_2^2 \quad \text{effective inertia of } \theta_2 \quad (7)$$

$$e_1 = (m_1 l_1 + m_2 l_3) g \quad \text{gravity torque on rod 1} \quad (8)$$

$$e_2 = m_2 l_2 g \quad \text{gravity torque on rod 2} \quad (9)$$

The equations of motion, with friction collected on the right:

$$\begin{pmatrix} a & b \cos(\theta_1 - \theta_2) \\ b \cos(\theta_1 - \theta_2) & d \end{pmatrix} \begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} = \begin{pmatrix} -b \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - e_1 \sin \theta_1 + Q_1 \\ +b \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - e_2 \sin \theta_2 + Q_2 \end{pmatrix} \quad (10)$$

Friction takes the simplest Rayleigh form: each bearing resists the rate at which it is actually being turned. The pivot resists  $\dot{\theta}_1$ , the axle resists the *relative* rate  $\dot{\theta}_2 - \dot{\theta}_1$ :

$$\mathcal{F} = \frac{1}{2} c_1 \dot{\theta}_1^2 + \frac{1}{2} c_2 (\dot{\theta}_2 - \dot{\theta}_1)^2, \quad Q_i = -\frac{\partial \mathcal{F}}{\partial \dot{\theta}_i} \quad (11)$$

$$Q_1 = -(c_1 + c_2) \dot{\theta}_1 + c_2 \dot{\theta}_2, \quad Q_2 = c_2 \dot{\theta}_1 - c_2 \dot{\theta}_2 \quad (12)$$

Energy, with zero taken at both rods hanging at rest:

$$E = \frac{1}{2} a \dot{\theta}_1^2 + \frac{1}{2} d \dot{\theta}_2^2 + b \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + e_1 (1 - \cos \theta_1) + e_2 (1 - \cos \theta_2) \quad (13)$$

Two predictions follow at once, and both are used as checks. The small-oscillation frequencies of the linearised system should be 0.763 Hz and 1.830 Hz. And a flip of rod 2 needs enough energy to lift its centre of mass over the axle,

$$E > 2e_2 = 2.02 \text{ J}, \quad \text{which on the line } \theta_{2,0} = 0 \text{ means } \theta_{1,0} > 87.3^\circ. \quad (14)$$

Below that angle a flip is impossible whatever the dynamics does.

### 4 What the model predicts

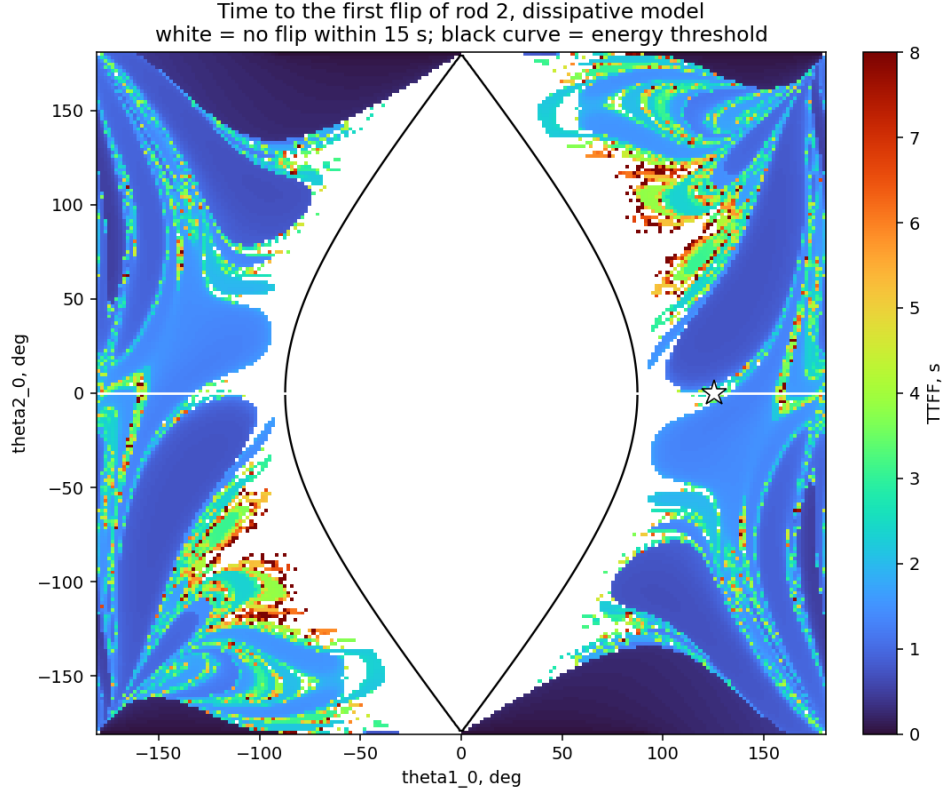


Figure 2: Time to the first flip over all pairs of release angles, dissipative model. White means no flip within 15 s. The black curve is the energy threshold (14); it lands exactly on the edge of the central white region, a good check that the map is computed correctly. The white horizontal line is our experiment,  $\theta_{2,0} = 0$ ; the star marks the predicted island. Friction widens the white region well beyond the black curve — a frictionless map has colour there.

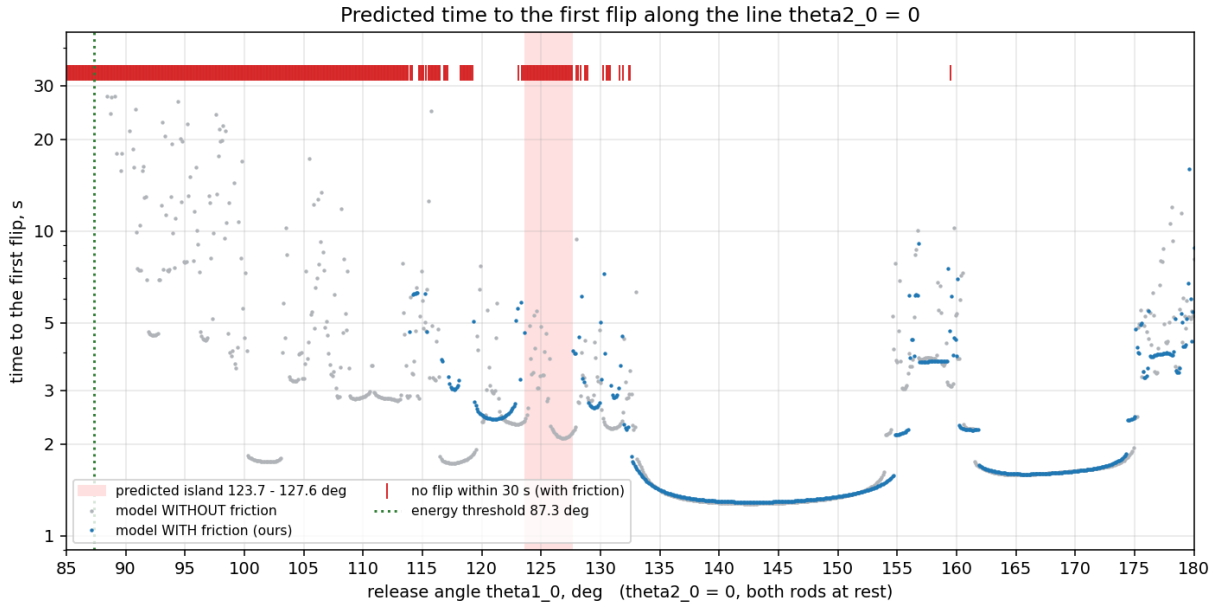


Figure 3: The same map read along our line. Grey is the frictionless model, blue is ours; red ticks mark angles that never flip within 30 s. Below  $87.3^\circ$  nothing can flip, above  $133^\circ$  the flip is fast and reliable. The shaded band at  $123.7^\circ$ – $127.6^\circ$  is the island: our model says no flip in 30 s, the frictionless model says a flip in 2–5 s.

## 5 Plan — three sessions of six hours

### Session 1 — take it apart and measure the pieces

Nothing here involves the double pendulum. Each rod is measured alone.

- 0:00–0:30 **Set up.** Camera on a tripod, square to the plane of motion, 120 fps. A ruler in the frame for the scale. Even light, no flicker.
- 0:30–1:30 **Geometry and mass.** Weigh each rod with its bearing hardware. Find the centre of mass by balancing on a knife edge and measure  $l_1$ ,  $l_2$  from the hinge; measure  $l_3$  between hinge centres. Three repeats of every length.
- 1:30–3:00 **Periods.** Hang rod 1 alone from the top pivot, release from about  $5^\circ$ , time 20 swings, five repeats. Same for rod 2 on its own axle. Small angles matter: at  $20^\circ$  the period is already 1% long.
- 3:00–4:30 **Decay.** Record a long free decay of each rod (60 s if the camera allows, otherwise two takes). Track it, plot the amplitude envelope, read the time to half amplitude. Check that the envelope really is exponential — if it falls in a straight line instead, there is dry friction and we should say so.
- 4:30–6:00 **Numbers into the model.** Compute  $I_1$ ,  $I_2$ ,  $\tau_1$ ,  $\tau_2$ ,  $c_1$ ,  $c_2$ . Enter mass, length, period and half-life into the Settings tab of `dp_sim.py`, which does the conversion itself and shows the derived constants. Write down the predicted small-oscillation frequencies.

### Session 2 — assemble, check the model, calibrate the top bearing

This session decides whether the model can be trusted at all. If the frequencies are wrong, no prediction about chaos is worth making.

- 0:00–1:00 **Small oscillations.** Release from about  $5^\circ$ , record 30 s, take the two frequencies from the spectrum of  $\theta_1(t)$ . They should match 0.763 Hz and 1.830 Hz to a few percent. A larger error means a length or an inertia is wrong — fix it before going on.
- 1:00–2:00 **Free decay of the assembled pendulum.** From about  $20^\circ$ , record 30 s, compute the energy from the tracked angles and fit its decay. Expect roughly a factor of two over 30 s.
- 2:00–3:00 **Calibrate  $c_1$ .** The top bearing now carries both rods, so its friction is larger than measured on rod 1 alone. Increase  $c_1$  (that is, shorten  $\tau_1$ ) until the simulated energy decay matches the measured one. Change nothing else. Report the factor — it is a result in itself.
- 3:00–4:30 **Trajectory check.** Release from  $60^\circ$  and  $90^\circ$ , three takes each. Overlay measured and simulated  $\theta_2(t)$ . They should agree for the first 3–5 s and then drift apart. That drift is not a failure: it is the chaos we came for.
- 4:30–6:00 **Rehearse the release.** Build and test the release: a fixed angle template, a hook or magnet, no push. Practise until repeated releases from the same setting agree to better than  $1^\circ$ , and rod 2 is hanging still at the moment of release. Measure that scatter — it sets how narrow an island we can hope to resolve.

### Session 3 — the hunt

One long sweep of release angles, then a fine sweep across the predicted island.

- 0:00–0:30    **Re-check.** One small-amplitude take to confirm nothing shifted overnight, and one release at  $150^\circ$ , where a fast flip is certain, to confirm the whole chain still works.
- 0:30–2:30    **Coarse sweep.** Release angles from  $80^\circ$  to  $170^\circ$  in steps of  $10^\circ$ , three takes each: 30 recordings. Read the TTFF straight off each video — a flip is obvious, no processing needed. Plot TTFF against angle beside the prediction.
- 2:30–5:00    **Fine sweep across the island.** From  $118^\circ$  to  $134^\circ$  in steps of  $1^\circ$ , five takes each: 85 recordings, about 90 s of work apiece. Record every outcome, including the ones that flip immediately.
- 5:00–6:00    **Wrap up.** Plot the measured TTFF.

## 6 How we judge the result

The comparison is not a curve fit. At a given angle chaos makes the individual TTFF value irreproducible on purpose; what should reproduce is the *statistics* — whether flips happen at all, and roughly how soon.

| Claim            | Passes if                                                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy threshold | No flip is ever seen below $87^\circ$ , in any take.                                                                                                                                                               |
| Fast-flip region | Above $135^\circ$ every take flips, and the measured TTFF sits within about 0.3 s of the predicted 1.3–1.7 s.                                                                                                      |
| The island       | Inside $124^\circ$ – $128^\circ$ the no-flip fraction is clearly higher than at the angles on either side. With five takes per angle, three or more no-flips inside against none outside is already a real signal. |
| Friction matters | The behaviour near the island follows the dissipative model, not the frictionless one, which predicts a flip within 5 s there.                                                                                     |

**Expect this to be hard.** The island is about  $4^\circ$  wide and its edges are fractal, so a release scatter of  $\pm 1^\circ$  blurs it badly. If it is not found, that is a result too — but then check the obvious causes first: the release repeatability measured in session 2, whether  $\theta_{2,0}$  was truly zero at release, and whether the bearings have warmed up and changed  $c_1$  since the calibration. Re-run the prediction with the calibrated  $c_1$  before session 3: the island moves when friction changes, and that is exactly what makes it a friction measurement in disguise.

## 7 What to hand in

1. A parameter table: every measured quantity with its scatter, and the derived  $I_1$ ,  $I_2$ ,  $c_1$ ,  $c_2$ ,  $a$ ,  $b$ ,  $d$ ,  $e_1$ ,  $e_2$ .
2. Predicted versus measured small-oscillation frequencies, and the calibration factor applied to  $c_1$ .
3. One overlay of measured and simulated  $\theta_2(t)$  at a medium angle, showing where they part.
4. The measured TTFF against release angle, with the model curve behind it.
5. A short answer to the question the project asks: is there an island, where is it, how wide, and does friction explain it.

All predictions in this document use  $m_1 = 0.438$  kg,  $m_2 = 0.480$  kg,  $l_1 = l_2 = 0.215$  m,  $l_3 = 0.255$  m,  $T_1 = T_2 = 1.00$  s and a 30 s half-amplitude time for both rods.